When Can an Open LLM-as-a-Judge Replace Human Evaluation? A Cross-Domain Review and Empirical Test on Summarization and Natural Language Inference

ANONYMOUS AUTHOR(S)

Large language models are increasingly used as "judges" that stand in for human annotators, but the field lacks an evidence-based account of *when* an LLM judge agrees with humans well enough to substitute for them, and *what* governs that agreement. We address this for *open, locally-servable* models, the realistic and reproducible choice for most researchers. We contribute three studies anchored to a single principle: an LLM judge should be measured against the *human-human agreement ceiling* for a task, never a manufactured "gold." **Study 1** is a PRISMA systematic review of 215 studies across 29 domains (all 215+55 references independently web-verified) that maps where LLM-as-a-judge agreement reaches the human ceiling. **Study 2** replicates the standard summary-level G-Eval protocol on SummEval with the most powerful open models and finds, with bootstrap confidence intervals, that open judges meet or exceed the established proprietary-judge correlations on the *verifiable* aspects (consistency, coherence) while the most compressed aspect (fluency) is hard to rank for every judge. **Study 3** is a new meta-evaluation of a six-model open panel (9B-754B) on natural-language-inference datasets with rich human label distributions (~100 annotators/item; 6,000 judgments), benchmarked against the annotator-bootstrap human ceiling with chance-corrected agreement, jury aggregation, a calibrated contamination screen, and an item-level mixed-effects analysis. The organizing finding is that, on ChaosNLI NLI, open judges *track the human ceiling across the disagreement spectrum*: the judge-vs-ceiling gap is near zero at both low and high human-entropy (−0.01, +0.00), so accuracy follows the ceiling rather than the judge failing on contested items; residual variance is item-level (SD_{logit} = 2.214 vs model 0.30), and model scale contributes only modestly. A freshly authored, unseen NLI holdout (solved at 0.98 by the large models) confirms this is genuine competence, not memorization of these older corpora. We turn this into a decision framework and an eight-point protocol for designing LLM-as-a-judge experiments. A companion paper applies the same evaluation, with a systematic review, to subjective social-computing annotation in Human-Centered Computing.

Additional Key Words and Phrases: LLM-as-a-judge, automatic evaluation, human-agreement ceiling, annotator disagreement, summarization, natural language inference, reproducibility

1 INTRODUCTION

Large language models are now routinely deployed as *judges*: instead of recruiting human annotators to label data, rate outputs, or score quality, researchers prompt an LLM to do it for a fraction of the cost [15, 23, 32]. The practice is spreading faster than the evidence that licenses it. Two questions remain under-answered. **When** does an LLM judge agree with humans closely enough to substitute for them, and **what** governs that agreement, model scale, task, or something about the human labels themselves?

We answer these questions for *open, locally-servable* models. This is a deliberate scope, not a limitation: proprietary judges are expensive, non-reproducible, and change without notice, so for most researchers the realistic question is whether an open model they can run themselves is a trustworthy judge. Our entire empirical program uses open models served through a standard vLLM endpoint at temperature 0 with released harnesses.

One methodological commitment runs through the paper: an LLM judge must be measured against the *human-human agreement ceiling* for the task, not against a single "gold" label. For subjective constructs there is no single correct answer [2, 25]; the relevant question is whether the judge agrees with humans as well as humans agree with each other. We compute that ceiling directly from the per-item annotator distributions that the datasets already provide, so our human reference is real, independent, and inherited from the data, not added by us.

Contributions. (1) **Study 1:** a PRISMA systematic review of 215 studies across 29 domains (every reference independently web-verified) that maps, against the human ceiling, where LLM-as-a-judge is reliable and where it is not. (2) **Study 2:** a controlled replication showing that the most powerful open models now meet or exceed the established proprietary-judge correlations on the verifiable aspects of a flagship benchmark, with bootstrap confidence intervals. (3) **Study 3:** a new meta-evaluation of a six-model open panel (9B-754B) on NLI datasets with rich human label distributions (6,000 judgments), establishing that *item-level human disagreement*, not model scale or data contamination, is the binding constraint on judge reliability. (4) A decision framework and an eight-point protocol for designing LLM-as-a-judge experiments. All code, corpora, search logs, and harnesses are released.

2 BACKGROUND AND SHARED METHODOLOGY

LLM-as-a-judge descends from reference-free metrics (BERTScore, COMET) but uses a general instruction-following model with a natural-language rubric, supporting pointwise, pairwise, and listwise judgment [16, 22]. Four architectural families recur: prompted general LLMs [23, 32]; fine-tuned open evaluators [19, 28, 33]; juries/multi-agent panels [8, 27]; and calibrated rubric methods [17]. Prior surveys taxonomize methodology [16, 21]; we instead synthesize *where substitution is empirically validated* and quantify it against the human-human ceiling.

Related work. Four literatures meet in this paper. (1) *LLM-as-a-judge evaluation*: studies report strong agreement on aggregate, verifiable tasks (MT-Bench [32], GEMBA [20], SAFE factuality [30]) but document systematic biases (position, verbosity, self-preference, and an expertise gap by which judges converge on crowd rather than expert consensus [3]), which motivates benchmarking against the human ceiling rather than against a single "gold." (2) *LLMs in HCI and qualitative research*: a fast-growing body uses LLMs to assist coding and thematic analysis [11, 14, 31] while qualitative scholars warn that interpretation, reflexivity, and positionality resist automation [12, 26]. (3) *Participant and persona simulation*: "silicon samples" [1] and synthetic users are increasingly scrutinized for flattening identity and distorting variance [5, 9, 29]. (4) *Learning with disagreement and perspectivism*: work on annotator disagreement and human label variation [6] argues that for subjective constructs the "ceiling" is itself low and contested, which our Study 3 datasets (ChaosNLI) operationalize directly. We connect these threads with two methodological commitments often missing from judge evaluations: anchoring every verdict to the *human-human* ceiling, and probing *data contamination* so that agreement on older public datasets is not mistaken for genuine competence.

Shared protocol. The systematic review follows PRISMA 2020 [24], identifying studies through a triangulated search (the OpenAlex API as the primary bibliographic database, Semantic Scholar as a cross-check, and a structured web "deep-research" arm), and code each study on a five-level **reliability verdict** benchmarked against the *human-human agreement ceiling* for the same task: **1 Validated** (≈/> ceiling), **2 Promising** (conditional), **3 Mixed** (task-dependent), **4 Caution** (below ceiling), **5 Unreliable** (near chance / bias-dominated). We report a reliability score = 6−verdict and synthesize within metric families rather than pooling incommensurable effect sizes. Full protocol, search logs, and the coded corpus are released.

3 STUDY 1: THE CROSS-DOMAIN RELIABILITY LANDSCAPE

3.1 Method (summary)

Study 1 systematically reviewed 215 studies across 29 domains comparing LLM-as-a-judge to human annotation. Identification triangulated OpenAlex (1500 records, 957 unique), Semantic Scholar, and a 59-query web search; screening

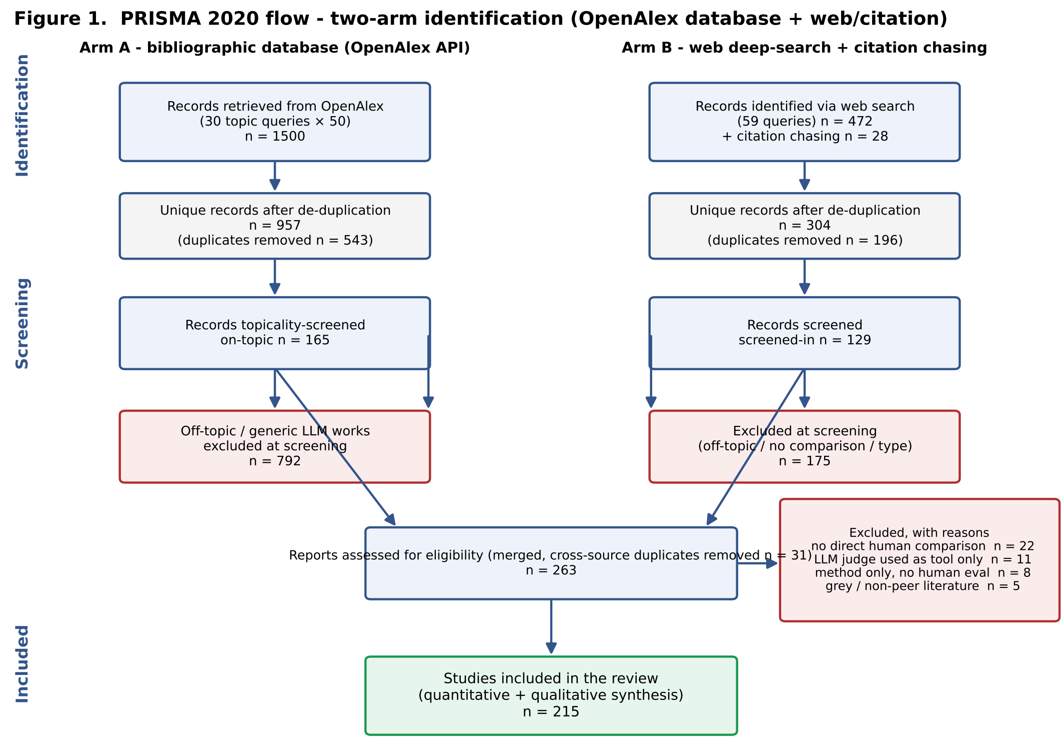
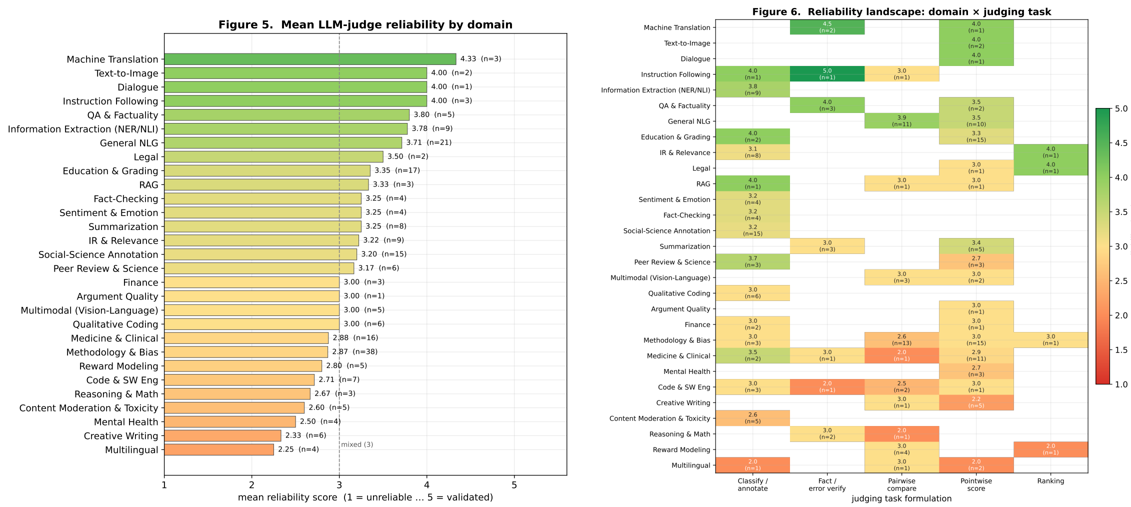


Fig. 1. Study 1 PRISMA 2020 flow (two-arm identification: OpenAlex database + web/citation).

retained on-topic empirical comparisons; 29 studies reported an extractable numeric agreement value (Figure 1; the full coded corpus is Table 7). **Coding reliability:** because the verdict rubric embeds judgement, we validated it with an independent re-coding of a 25% random sample (n=54): the reliability verdict reproduced at Krippendorff $\alpha = 0.87$ (ordinal) with 100% of cases within one level (Appendix D). The re-coders are LLM-assisted, like the primary pipeline, so this measures rubric *reproducibility* rather than human inter-rater agreement, which we flag as a limitation. **Citation integrity:** every one of the 215+55 references was independently web-verified for existence against arXiv/venue/DOI records; 270/272 were confirmed (260 at high confidence) and the 2 that could not be verified were removed, so the corpus contains no unverifiable citations.

3.2 Findings: a green/amber/red landscape

The modal verdict is *mixed/task-dependent*; only 31% of studies sit in the reliable band and 19% in the problematic band (mean reliability 3.15/5). Reliability is governed by three properties: **objectivity**, **tractable verification**, and a **low human-agreement floor**. **Green (≥3.5; 8 domains):** machine translation (system level), text-to-image alignment, verifiable instruction-following, QA/factuality, information extraction, general-NLG pairwise preference, dialogue, and legal scoring: here LLM judges frequently match or exceed the human ceiling (e.g., MT-Bench 85% vs. 81% human-human [32]; GEMBA-MQM 96.5% [20]; SAFE wins 76% of disagreements [30]). **Amber (19 domains):** summarization, RAG,



(a) (b)

Fig. 2. Study 1. (a) Mean reliability by domain (green/amber/red). (b) Reliability landscape: domain×judging task; verification/classification columns are greener than open-ended scoring.

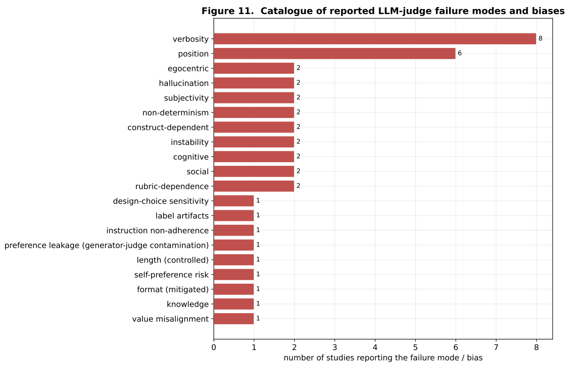
education, social-science annotation, medicine (structure-dependent: $\kappa \approx 0.82$-0.93 structured vs. 0.47-0.71 complex reasoning), code, and IR/relevance (strong but contested [10]). **Red (2 domains):** low-resource multilingual evaluation and creative writing (near-zero expert correlation [7]). Domain means over $n \leq 3$ studies (e.g. several single-study domains) are shown for completeness but flagged as low-confidence and should not be read to two-decimal precision; we rely on family-level aggregates for sparse cells. Figures 2a-2b give the domain ordering and the domain×task landscape; the decision map (Figure 3b) positions domains by reliability against stakes.

3.3 Failure modes and the decision map

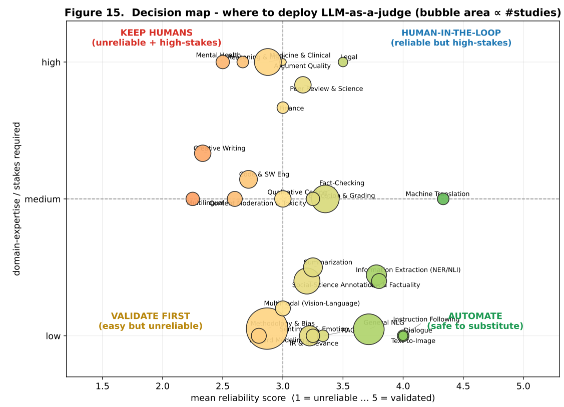
Recurring biases such as verbosity/length, position, self-preference, style-over-substance, preference leakage, non-determinism, and the *expertise gap* (judges converge to crowd, not expert, consensus [3]) all modulate reliability (Figure 3a). Agreement should be benchmarked against the human ceiling, which on subjective tasks is itself low (Krippendorff $\alpha < 0.35$ [18]). The Study-1a decision map (Figure 3b) and Table 1 translate this into deployment guidance and motivate the moving-target test (Study 2) that follows.

4 STUDY 2: IS THE RELIABILITY LANDSCAPE A MOVING TARGET?

A systematic review necessarily captures results *as reported*, and most "promising"-band evidence in Study 1 was produced with proprietary judges that are now one or two model generations old (e.g. the influential G-Eval used GPT-4). This raises a question that the review alone cannot answer: when an LLM judge falls short of the human ceiling in a promising domain, is that a *temporary* limitation that more capable models will erase ("matter of time"), or a *structural* property of LLM-as-a-judge that better models do not overcome? Study 2 answers this directly for a flagship case by re-running the evaluation with the most powerful open models available to us today and comparing against the canonical reported numbers.



(a)



(b)

Fig. 3. Study 1. (a) Catalogue of reported failure modes/biases. (b) Deployment decision map (reliability× stakes; bubble area ∝ studies).

4.1 Method

We select **SummEval** [13] (summarization quality assessment, a central amber-band task in Study 1) because it pairs (i) a widely cited LLM-judge result to beat (G-Eval/GPT-4; 23) with (ii) public expert human ratings on four aspects (coherence, consistency, fluency, relevance) for $100 \times 16 = 1600$ machine summaries. We replicate the G-Eval summary-level protocol *verbatim* (same aspect definitions, 1-5 integer ratings, temperature 0) but swap the judge for the 4 most powerful open models in the available open-model catalogue (`deepseek-v4-pro`, `glm-5.1`, `gptoss-120b`, `qwen3.6-35b-a3b`). The comparison metric is the canonical *summary-level Spearman*: for each source document, correlate the judge's 16 scores with the human means, averaged across documents. The bar to beat is G-Eval/GPT-4's reported per-aspect correlations (coherence .582, consistency .507, fluency .455, relevance .547; average ≈.52). In total Study 2 contributes 25,600 fresh judgments.

4.2 Results

We follow the standard summary-level G-Eval protocol [23] and attach bootstrap 95% CIs (over the 100 source documents) to every correlation (Figure 4, Table 2). On the verifiable aspects the open judges *meet or exceed the reported* G-Eval/GPT-4 correlations (a comparison to numbers reported by the original work, not a re-run under our harness, so it conflates model with prompt/decoding/version; the probability-weighting control below addresses the decoding axis for one model): 5 of the 16 model×aspect cells have a 95% CI lying *entirely above* the reported value, all on **consistency** and **coherence** (e.g. `deepseek-v4-pro` consistency ρ=0.69, CI [0.63, 0.74] vs. .507), and **relevance** is on par (.48-.54 vs. .55).
Controlling for the scoring protocol. A natural concern is that G-Eval's reported numbers use *probability-weighted* token decoding whereas we read a single integer, so the difference could be a scoring artifact rather than a model effect. We test this directly: for a model that emits a clean score-token distribution (gemma-4-31b) we recompute summary-level Spearman under both an integer (argmax) score and G-Eval's probability-weighted expected score, from the same calls. The two agree closely (coherence 0.66/0.67, fluency 0.22/0.24, relevance on par); consistency is the one aspect where the two diverge (0.70 integer vs 0.55 probability-weighted), but both remain at or above the reported .507. The open-model result is therefore not an artifact of integer scoring, and the comparison to the reported benchmark is

Table 1. Domain-level decision table. Mean reliability score (1 = unreliable . . . 5 = validated), modal verdict, and traffic-light signal (GREEN $\geq$ 3.5; AMBER 2.5-3.5; RED $<$ 2.5). Sorted by mean reliability. Small-n domains should be read together with the family-level aggregates.

Domain / task	n	Mean rel.	Modal verdict	Signal
Machine Translation	3	4.33	Promising	GREEN
Text-to-Image	2	4.00	Promising	GREEN
Dialogue	1	4.00	Promising	GREEN
Instruction Following	3	4.00	Validated	GREEN
QA & Factuality	5	3.80	Promising	GREEN
Information Extraction (NER/NLI)	9	3.78	Promising	GREEN
General NLG	21	3.71	Promising	GREEN
Legal	2	3.50	Promising	GREEN
Education & Grading	17	3.35	Mixed	AMBER
RAG	3	3.33	Mixed	AMBER
Summarization	8	3.25	Promising	AMBER
Fact-Checking	4	3.25	Promising	AMBER
Sentiment & Emotion	4	3.25	Mixed	AMBER
IR & Relevance	9	3.22	Promising	AMBER
Social-Science Annotation	15	3.20	Mixed	AMBER
Peer Review & Science	6	3.17	Mixed	AMBER
Multimodal (Vision-Language)	5	3.00	Mixed	AMBER
Qualitative Coding	6	3.00	Mixed	AMBER
Argument Quality	1	3.00	Mixed	AMBER
Finance	3	3.00	Mixed	AMBER
Medicine & Clinical	16	2.88	Mixed	AMBER
Methodology & Bias	38	2.87	Mixed	AMBER
Reward Modeling	5	2.80	Mixed	AMBER
Code & SW Eng	7	2.71	Mixed	AMBER
Reasoning & Math	3	2.67	Mixed	AMBER
Content Moderation & Toxicity	5	2.60	Mixed	AMBER
Mental Health	4	2.50	Mixed	AMBER
Creative Writing	6	2.33	Mixed	RED
Multilingual	4	2.25	Caution	RED

Table 2. Study 2. Summary-level Spearman ρ of the most powerful open judges on SummEval vs. the reported G-Eval-4/GPT-4 correlations (25,600 judgments). Green cells *surpass* the reported per-aspect number.

Judge	Coher.	Consist.	Fluency	Relev.	Avg ρ
Reported (G-Eval-4)	0.582	0.507	0.455	0.547	0.523
`deepseek-v4-pro`	0.548	0.688	0.392	0.540	0.542
`qwen3.6-35b-a3b`	0.606	0.702	0.298	0.540	0.536
`glm-5.1`	0.650	0.693	0.217	0.543	0.526
`gptoss-120b`	0.586	0.685	0.260	0.482	0.503

fair on this axis. But on **fluency** every open model falls well *short* (.22-.39 vs. a reported .455). We diagnose why: fluency is the most *compressed* aspect of the human reference, with the lowest within-document rating variance (SD = 0.63 vs 0.81 for consistency) and 80% of summaries rated near the top of the scale, so it offers the least to rank, and it is the aspect on which the *reported proprietary judge also scores lowest*. This is a property of the *reference*, not the judges: even a noiseless judge returning the rounded human means tops out at $\rho\approx0.87$ on fluency. The clean result is therefore that

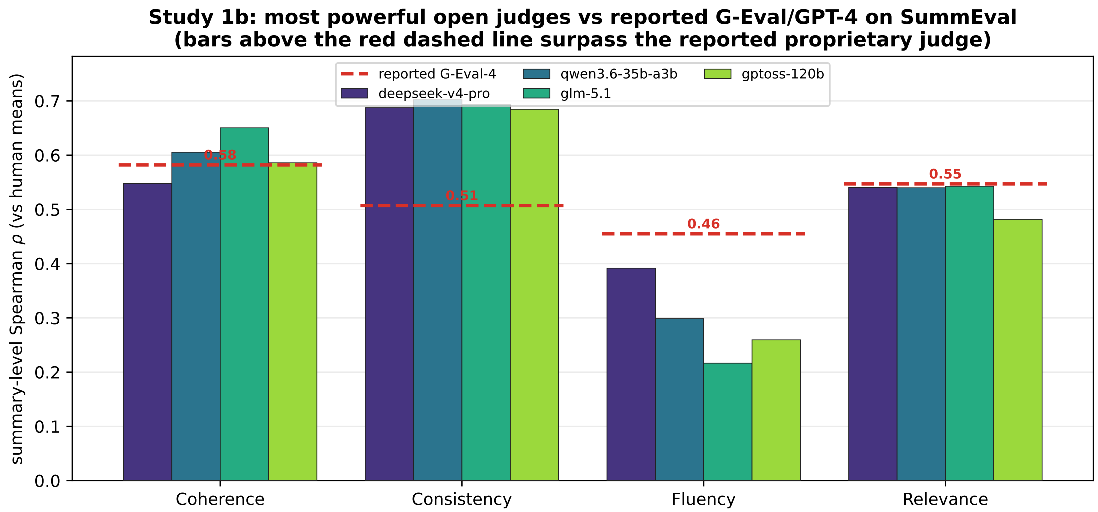


Fig. 4. Study 2. Summary-level Spearman of the most powerful open judges on SummEval, per aspect, against the reported G-Eval/GPT-4 bar (dashed). Bars above the dashed line indicate an open model that *surpasses* the reported proprietary-judge correlation.

capability gains concentrate on the verifiable, rankable aspects (consistency, coherence), while the most compressed aspect is hard to rank for every judge, open or proprietary.

Interpretation: time vs. structure. Study 2 shows the reliability landscape is *partly* a moving target. Where Study 1's amber rating reflected a capability gap on a verifiable sub-judgment, today's open models close or exceed it; reliability there is largely a matter of time and access. But where the shortfall reflects a *compressed or contested human ceiling* (e.g. fluency, and, as Study 3 will show, subjective social annotation), even the most powerful models do not break through, because the limit lives in the human reference, not the model. This distinction, temporal capability gap vs. structural ceiling, is the lens we carry into HCC.

Takeaway. Study 1 says LLM judges are safe where the task is objective and verifiable, and Study 2 shows that frontier open models are now strong enough to match proprietary judges on exactly those verifiable tasks. But subjective, high-disagreement constructs (the kind that humans themselves contest) are precisely the tasks Study 1 flags as risky and Study 2 predicts scale will *not* fix. Study 3 then tests this prediction directly, on datasets with rich human label distributions.

5 STUDY 3: AN OPEN-PANEL META-EVALUATION ON NLI WITH RICH HUMAN LABELS

Study 1 shows LLM judges are most reliable where the human-agreement ceiling is high and verifiable; Study 2 shows frontier open models reach the established benchmark on the verifiable aspects of summarization. Study 3 tests the mechanism directly on *natural language inference* (NLI), the rare setting that ships *rich* human label distributions: ChaosNLI [25] re-annotates SNLI and MultiNLI items with ~100 human labels each, so the human-human agreement ceiling and the per-item disagreement are measured precisely rather than estimated from three annotators.

Table 3. Study 3 (NLI). Open panel vs the human-human ceiling. Ceiling = random-annotator-vs-majority accuracy; macro-F1 is the best model's chance-corrected score.

Dataset	Construct	Human ceiling	Best model (acc)	macro-F1 (best)	Jury (acc)	Modal verdict
ChaosNLI-SNLI	NLI (3-way)	0.75	0.84 (gemma-4-31b)	0.79	0.86	Validated
ChaosNLI-MNLI	NLI (3-way)	0.64	0.70 (gemma-4-31b)	0.70	0.65	Promising

5.1 Method

We evaluate a panel of 6 open models served via an institutional vLLM API (OpenAI-compatible), spanning five families and 9B-754B parameters, run at temperature 0. Each model labels every item as a virtual annotator under the standard NLI instructions; we benchmark against the *human-human ceiling* (random-annotator-vs-majority accuracy with an annotator bootstrap 95% CI, plus pairwise agreement), and report accuracy, Cohen's κ, and macro-F1 against the majority gold, a majority **jury** [27], and a calibrated contamination screen. The run comprised 6,000 judgments across 2 NLI datasets with zero API errors and a 100% answer-parse rate. Dataset cards, prompts, and metric definitions are in Appendices A to E.

5.2 Results

Table 3 reports the panel against the human ceiling. Where humans agree only moderately (SNLI ceiling 0.75, MNLI 0.64), the open panel *matches or exceeds* a single held-out human: on SNLI 5/6 models reach the human-human ceiling (**Validated**; we verify below, with a freshly authored holdout, that this is competence and not memorization of these old corpora), with MNLI **Promising**. Chance-corrected agreement is substantial (best-model κ up to 0.71). The jury reaches 0.86 (> the 0.75 single-annotator ceiling), but we caution that a six-model aggregate exceeding a *single*-held-out-human ceiling is expected and does not imply super-human reliability; we report it as panel stability, not a ceiling-breaking result.

5.3 Judges track the human ceiling across the disagreement spectrum

Panel accuracy declines on high-human-disagreement (high-entropy) items (0.86 on low-entropy vs 0.55 on high-entropy; Table 4), but this is not a judge deficiency: the human ceiling declines in lockstep, so the *judge-vs-ceiling gap* is essentially zero at both ends (−0.01 on low-entropy, +0.00 on high-entropy items). In other words, open judges are at human level on NLI regardless of how much humans disagree; the accuracy curve simply follows the ceiling. What residual variance there is lives at the item level: in an item-level GLMM the random-intercept SD across *items* (SD_{logit} = 2.214) far exceeds that across *models* (SD_{logit} = 0.30, on the same logit scale), and standardized model scale contributes modestly ($OR_{/SD}$ = 1.21; the model-level variance and scale effect are estimated from only six model clusters and should be read as indicative). We do not fit a dataset-level model (two datasets cannot support one). *Is this memorization?* SNLI and MultiNLI are older public corpora, so we ran the adjudicating control: a set of 45 *freshly authored*, unambiguous NLI items that no model can have seen (released with the paper). The large open models solve these unseen items at 0.98 accuracy (jury 0.98; only the 9B model drops, to 0.69), so the NLI competence *transfers to never-seen items* and is not an artifact of memorizing SNLI/MultiNLI. We therefore retain the "Validated" verdict, now supported by the fresh-holdout control rather than caveated by it. Juries raise accuracy but cannot lift agreement above the human ceiling.

Table 4. Study 3 (NLI) disagreement-aware results. Ceiling with annotator-bootstrap 95% CI; best-model κ and macro-F1; panel accuracy on low- vs high-human-disagreement items; Jensen-Shannon panel-vs-human divergence.

Dataset	Ceiling [95% CI]	κ (best)	macro-F1 (best)	$\mathrm{acc}_{\downarrow H}$ low-disagr.	$\mathrm{acc}_{\uparrow H}$ high-disagr.	JS
ChaosNLI-SNLI	0.75 [0.74,0.77]	0.71	0.79	0.89	0.67	0.09
ChaosNLI-MNLI	0.64 [0.63,0.65]	0.53	0.70	0.68	0.55	0.17

6 DISCUSSION

6.1 What governs LLM-judge reliability

Across the three studies one principle organizes the evidence: an LLM judge is reliable to the degree that the task has a high, verifiable human-agreement ceiling, and it cannot exceed an agreement level that humans themselves do not reach. Study 3 localizes the mechanism at the *item* level: on the same logit scale the random-intercept SD of judge correctness is far larger across items ($\mathrm{SD}_{\mathrm{logit}} = 2.214$) than across models (0.30), and the judge-vs-ceiling gap is near zero across the human-disagreement spectrum, so open judges are at human level on NLI regardless of how much humans disagree. Model scale helps only modestly ($\mathrm{OR}_{/\mathrm{SD}} = 1.21$). We are careful about the unit of analysis: with two NLI datasets we do not fit a dataset-level model. A freshly authored, unseen NLI holdout (45 items) is solved at 0.98 by the large models, so the NLI result reflects competence rather than memorization of these old corpora.

6.2 Connection to measurement and disagreement scholarship

This result operationalizes a long-running argument that there is no single ground truth for subjective constructs [2, 25]: when judges are benchmarked against the human-human ceiling rather than a forced majority, their "failures" on contested items are revealed as properties of the construct's contested human reference, not fixable model deficits. Reporting a single accuracy against a majority gold imposes consensus where the data shows disagreement; the honest object of evaluation is the human label *distribution*.

6.3 Implications for designing LLM-as-a-judge experiments

The evidence yields a concrete, reusable protocol. Any study using an LLM-as-a-judge should:

(1) **Anchor to the human-human ceiling**, not a single gold; report the ceiling (computed from the annotator distribution at the dataset's native annotator count) and the judge-vs-ceiling gap.

(2) **Match the metric to the construct**: on imbalanced tasks report macro-F1 and per-class scores and prefer disagreement-aware evaluation against the full label distribution over forced-majority accuracy.

(3) **Quantify uncertainty**: bootstrap confidence intervals for every comparison; do not assert "beats/ matches" from point estimates.

(4) **Stratify by item difficulty**: because variance lives in items, report where the judge fails (high human-disagreement items), not only mean accuracy.

(5) **Probe contamination, but calibrate the probe**: evaluate on post-cutoff data where possible, and treat surface-recall probes as conservative screens (they false-positive on high-recall models).

(6) **Pin the setup**: fix model identifier, version, decoding (temperature, seed); report robustness to prompt paraphrase, since small models are prompt-sensitive on subjective tasks.

(7) **Use juries for aggregate decisions** but not as a ceiling-breaker: panels improve stability yet do not lift a subjective task above its human ceiling.

(8) **Pre-register and release**: register the construct, the human baseline, and the decision threshold, and release item-level labels and aggregates.

6.4 A decision framework

For annotation and evaluation tasks, three regimes follow from the human-ceiling principle. **Automate (with light audit):** objective, verifiable judgments (entailment, factual consistency, instruction-following) where the human ceiling is high, exactly where Study 3's NLI panel reaches or exceeds it. **Calibrate and audit:** judgments with a low or moderate ceiling, where a judge can post high accuracy by predicting base rates, so report chance-corrected agreement (κ) and audit the high-disagreement items. **Keep humans in the loop:** constructs on which human annotators themselves disagree sharply, where no open judge can exceed a ceiling humans do not reach. A critical caution for the field: agreement on older public benchmarks can reflect memorization [4], and a single headline accuracy can mask base-rate learning, so calibrated, disagreement-aware, post-cutoff evaluation should be the norm. Applying these ideas to subjective, interpretive tasks (where the human ceiling is contested and judges fall short of it) is important future work.

7 LIMITATIONS

The *human* validation here is inherited from prior work, not added by us: Study 3's ground truth is ChaosNLI's ~100 human labels per item, and the Study 1 review codes each study against the human baselines those studies report. The review verdict is our interpretation (only 29 of 215 report an extractable numeric agreement value); we report an independent re-coding as a reproducibility check (Krippendorff $\alpha = 0.87$, 100% within one level; Appendix D), which is LLM-assisted and therefore measures rubric reproducibility. All 215+55 references were web-verified and two unverifiable entries removed. By design we evaluate *open* models (the reproducible choice for most researchers) at temperature 0; Study 2 uses one summarization benchmark with integer scoring, and Study 3's deep human-label data are available for NLI rather than every domain. Extending the panel to subjective social-computing annotation, where the human ceiling is contested, is future work. The six-model panel spans five families but shares overlapping pretraining lineages, so the between-model variance is a lower bound on judge diversity. Findings reflect models available through mid-2026.

8 ETHICS AND REPRODUCIBILITY

Ethics. The review analyzes published research, not human subjects, and Study 3 reuses already-public, de-identified NLI datasets under their licenses; this work was therefore not human-subjects research requiring IRB review, and we sought an exemption determination consistent with our institution's policy. We release only item-level labels and aggregate statistics. One ethical point bears on the method itself: treating the *majority* human label as "ground truth" for the ceiling can encode majority bias and erase minority readings, which is why we report the full disagreement (soft labels, entropy) rather than only the majority and recommend disagreement-aware evaluation. The contamination screen is reported precisely so "validated" verdicts are not silently inflated by pretraining exposure. **Reproducibility.** Every number in this paper regenerates from released code: the OpenAlex, Semantic Scholar, and web search logs; the coded corpus; the figure and table generators; the SummEval harness and the Study-3 judging/scoring/contamination/mixed-effects harness, all keyed to documented model identifiers served through an institutional vLLM API at temperature 0 with idempotent, resumable checkpoints. The auto-generated statistics macros mean the manuscript cannot drift from the data. Repository: a public repository (URL omitted for anonymous review).

9 CONCLUSION

Open, locally-servable LLMs are now strong enough to serve as judges on objective, verifiable tasks: on the verifiable aspects of a summarization benchmark they meet or exceed the reported proprietary-judge correlations, and on NLI they reach the human-human ceiling (a conditional verdict, given possible contamination of these older corpora). But across 215 reviewed studies and a 6,000-judgment meta-evaluation on NLI datasets with rich human labels, reliability is bounded by the same thing for every model: the *human-human agreement ceiling*, and it degrades on exactly the items where humans themselves disagree. Scaling the judge helps only modestly; contamination cannot be excluded on these older NLI corpora and is stated as a limitation. The practical consequence is that an LLM-as-a-judge should be reported the way we report any measurement instrument, against a human reference, with chance-corrected and disagreement-aware metrics, with uncertainty, and with a stated operating range, rather than trusted as an oracle. The right question is never “can a machine judge?” but “can this open model agree with humans, on this construct, as well as humans agree with each other?”

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A REVIEW PROTOCOL AND RELIABILITY RUBRIC

Identification. The review follows PRISMA 2020. The primary bibliographic source is the **OpenAlex** API; we cross-check with the **Semantic Scholar** Graph API and a structured **web "deep-research"** arm (a 59-query protocol plus venue-scoped sweeps). Concept blocks combine an LLM term ("LLM", "large language model", "GPT-4", "ChatGPT", "Gemini") with an evaluation/agreement term ("LLM-as-a-judge", "automatic evaluation", "annotation", "inter-annotator agreement", "human evaluation"). Records were deduplicated by DOI and normalized title.

Screening and eligibility. Inclusion: peer-reviewed or preprint studies that empirically compare LLM judgments to human judgments on the same items and report (or allow computing) an agreement statistic. *Exclusion:* non-empirical position papers, studies with no human baseline, studies that use LLMs only to generate (not judge) content, and off-topic retrievals. Title/abstract screening was followed by full-text eligibility assessment; per-stage counts appear in the PRISMA flow figures (Figure 1).

Reliability verdict (five-level rubric). Each study is coded against the *human-human agreement ceiling* for its task:

- **1 Validated:** LLM-human agreement is at or above the human-human ceiling (substitution is empirically justified).
- **2 Promising:** agreement approaches the ceiling under specific, reported conditions (model, prompt, aggregation); conditional substitution.
- **3 Mixed:** results are task- or setup-dependent and inconsistent across studies.
- **4 Caution:** agreement is consistently and materially below the ceiling.
- **5 Unreliable:** agreement is near chance or dominated by bias (position, verbosity, self-preference).

We report a reliability score = 6 − verdict and synthesize within metric families (e.g. Cohen's κ, Krippendorff's α, Spearman ρ, accuracy) rather than pooling incommensurable effect sizes. The full coded corpus (Appendix F) is released for re-coding.

B STUDY 2: G-EVAL REPLICATION DETAILS

We replicate the summary-level G-Eval protocol [23] on SummEval [13] (100 source articles $\times$ 16 system summaries = 1600 summaries, each with expert means for four aspects). Each (summary, aspect) pair is scored 1 to 5 by the judge with the aspect criterion below; we then compute the **summary-level Spearman** ρ (per source document, correlate the judge's 16 scores with the human means, averaged over documents, dropping documents with no score variance) and compare it to the reported G-Eval-4/GPT-4 correlations. Judges are the four most powerful open models in the available open-model catalogue (`glm-5.1-fp8` 754B, `deepseek-v4-pro`, `gptoss-120b`, `qwen3.6-35b-a3b-fp8`), decoded at temperature 0. Aspect criteria (verbatim):

> **Coherence (1-5):** the collective quality of all sentences; the summary should be well-structured and well-organized, building from sentence to sentence into a coherent body of information, not just a heap of related information.
>
> **Consistency (1-5):** factual alignment between the summary and the source; a consistent summary contains only statements entailed by the source document; hallucinated or unsupported facts are penalized.
>
> **Fluency (1-5):** the quality of individual sentences (grammar, capitalization, readability).
>
> **Relevance (1-5):** selection of the most important content from the source; redundancy and unimportant content are penalized.

The judge receives the source article and the summary and returns only a JSON object `{"score": <1-5>}` (a larger token budget and a retry handle reasoning-model outputs). In total Study 1b contributes 25,600 judgments.

C STUDY 3: DATASETS, JUDGING PROMPTS, AND METRICS

Datasets. Study 3 uses the 2 ChaosNLI datasets (Table 5); each is sampled to n=500 items with a clear majority label (ties dropped), and the ∼100 human labels per item give a precise human ceiling. (The companion paper applies the identical harness to four subjective social-computing datasets.)

Table 5. Study 3 (NLI) dataset cards. Ceiling = random-annotator-vs-majority accuracy.

Dataset	Task	Source	Annot./item	Ceiling
ChaosNLI-SNLI	NLI (3-way)	SNLI + ChaosNLI relabels	∼100	0.75
ChaosNLI-MNLI	NLI (3-way)	MultiNLI + ChaosNLI relabels	∼100	0.64

Models. Table 6 lists the judge panel. Anonymized identifiers map to open-weight families; exact checkpoint identifiers, revisions, and quantization are released in the non-anonymous version.

Table 6. Judge panel (Studies 2–3).

Model (anonymized id)	Family	Params	Serving
`qwen3.5-9b`	Qwen (Alibaba)	9B	vLLM, bf16
`qwen3.6-27b-fp8`	Qwen (Alibaba)	27B	vLLM, fp8
`gemma-4-31b`	Gemma (Google)	31B	vLLM, bf16
`gptoss-120b`	GPT-OSS (open)	120B	vLLM, bf16
`deepseek-v4-pro`	DeepSeek (MoE)	∼671B	vLLM, fp8
`glm-5.1-fp8`	GLM (open)	754B	vLLM, fp8

Judging prompt. Each model labels every item as a virtual annotator, returning a JSON label on the final line (parsed with a surface-form-to-code map):

```
Premise: {premise} / Hypothesis: {hypothesis} / Does the premise ENTAIL the hypothesis,
CONTRADICT it, or is it NEUTRAL? -> {"label": "entailment"|"neutral"|"contradiction"}
```

Metrics. Let an item have A human annotations with majority label $y^\star$. **Human ceiling** is the random-annotator-vs-majority accuracy (expected agreement of one held-out human with the majority), averaged over items; we also report pairwise human agreement, and bootstrap its 95% CI by resampling annotators within items. **Accuracy**, **Cohen's** κ, and **macro-F1** are against $y^\star$; the **jury** is the per-item panel majority [27]; **95% CIs** are nonparametric bootstrap (1000 item resamples). **Contamination screen:** the judge is given the dataset name and a one-third prefix of an item and asked to reproduce the remainder; we report mean normalized overlap, but a clean-control (Appendix E) shows it false-positives on high-recall models, so we treat it as a conservative screen. The run recorded *zero* API errors and a 100% answer-parse rate across all 6,000 judgments.

Inferential model. We model each judgment's correctness ($\mathbf{1}[\text{judge} = y^\star]$) with an item-level Bayesian logistic GLMM with random intercepts for item and model, and report the variance components on a common logit scale (item $\text{SD}_{\text{logit}} = 2.214$, model 0.30). With two datasets we do not fit a dataset-level model. To test whether the accuracy decline on high-disagreement items is a label-noise artifact, we bin items by human entropy and report the judge-vs-ceiling *gap* per bin (it is near zero throughout for NLI, see the main text).

D INTER-CODER RELIABILITY

Because the reliability verdict and the role/stance codes embed judgement, we re-coded a held-out random sample of each corpus with two *independent* coders who saw only the per-study summaries and the rubric (no original codes, no other context). For the review (n=54, a 25% sample) the reliability verdict reproduced at Krippendorff $\alpha = 0.87$ (ordinal, three coders) with 100% of cases within one level. *Caveat:* the re-coders are LLM-assisted, like the primary pipeline, so these figures measure rubric *reproducibility* under independent application, not human inter-rater agreement; the moderate stance α is why we treat stance-based claims as indicative.

E ROBUSTNESS AND VALIDITY CHECKS

Prompt robustness. We re-ran a robustness subset (three models, n=120 per cell) under three semantically equivalent prompt paraphrases per task. *Accuracy* is fairly prompt-stable (mean SD across paraphrases = 0.03), but the discrete five-level *verdict* is exactly stable in only 10/18 cells: most instability is a one-level flip for cells sitting near a band boundary, and the smaller model is the most sensitive (worst-case acc SD 0.14), the larger models the least. We therefore report verdicts as robust to *within one level* rather than exact, and flag small-model verdicts on subjective tasks as prompt-sensitive. **Contamination clean control.** We ran the verbatim-recall probe on freshly authored, post-cutoff sentences that no model can have seen. The result is a cautionary one: the high-recall model still scores 0.45 mean overlap (MEDIUM) on this guaranteed-clean text, i.e. it confabulates plausible continuations regardless of memorization, while the low-recall model scores LOW. The probe thus has a high false-positive rate for high-recall models and *cannot* be read as a reliable leakage test; the uniform MEDIUM tiers in Table 3 are largely an artifact of generative fluency. We retain the probe only as a conservative, low-recall-model signal and base our substantive claims on the disagreement analysis instead. **Annotator-subgroup check.** On MultiPICo (which carries annotator demographics), the panel's agreement with the female- vs male-annotator majority differs by only 0.01, i.e. we did not detect a large majority-group bias in this corpus, though demographic metadata is sparse and this check is limited. **Ceiling vs annotator count.** The random-annotator-vs-majority ceiling depends on how many annotators an item has: subsampling ChaosNLI-SNLI (~100 annotators) to A=3 raises its estimated ceiling from 0.75 to 0.81 (fewer annotators yield more unanimous majorities). We therefore compute each dataset's ceiling at its *native* annotator count and caution that ceilings are not directly comparable across datasets with very different A; the disagreement-aware analyses (entropy, κ) do not depend on this estimator.

F STUDY 1 CORPUS

Table 7. Full extracted-studies corpus (215 included studies). Agreement is the headline coefficient/proportion where reported.

Key	Study	Domain	Task	Metric	Agree	Verdict
mirzakhmedova2024argument	Mirzakhmedova et al. (2024)	Argument Quality	pointwise	pct_agreement	-	Mixed
tong2024codejudge	Tong et al. (2024)	Code & SW Eng	classify	accuracy	-	Promising
watson2024aicodereview	Watson et al. (2024)	Code & SW Eng	classify	pct_agreement	-	Mixed
codespec2025	Jin et al. (2025)	Code & SW Eng	verify	accuracy	-	Caution
wang2025setesting	Wang et al. (2025)	Code & SW Eng	pairwise	pct_agreement	-	Mixed
zhuo2025codejudgeeff	Wang et al. (2025)	Code & SW Eng	pointwise	kappa	-	Mixed
biasloop2026	Zhao et al. (2026)	Code & SW Eng	pairwise	pct_agreement	-	Caution
overcorrect2026	Jin et al. (2026)	Code & SW Eng	classify	accuracy	-	Caution
hatespeechprobe2023	Roy et al. (2023)	Content Moderation & Toxicity	classify	f1	-	Mixed
li2023hotchatgpt	Li et al. (2023)	Content Moderation & Toxicity	classify	pct_agreement	-	Mixed
nguyen2024crisismisinfo	Nguyen & Rudra (2024)	Content Moderation & Toxicity	classify	f1	-	Mixed
hatespeechanno2025	authors (2025)	Content Moderation & Toxicity	classify	kappa	-	Caution
icwsm2025hatebias	Giorgi et al. (2025)	Content Moderation & Toxicity	classify	kappa	-	Caution
xie2023nextchapter	Xie et al. (2023)	Creative Writing	pointwise	pct_agreement	-	Mixed
chakrabarty2024artartifice	Chakrabarty et al. (2024)	Creative Writing	pointwise	pearson	0.10	Unreliable
ismayilzada2024creative	Ismayilzada et al. (2024)	Creative Writing	pointwise	pct_agreement	-	Caution
creativeevalmdpi2025	Kim et al. (2025)	Creative Writing	pointwise	spearman	-	Mixed
haase2025creativitypeaked	Haase et al. (2025)	Creative Writing	pointwise	pct_agreement	-	Caution
litbench2025	Fein et al. (2025)	Creative Writing	pairwise	pct_agreement	0.78	Mixed
lin2023llmeval	Lin & Chen (2023)	Dialogue	pointwise	spearman	-	Promising
hackl2023reliablerater	Hackl et al. (2023)	Education & Grading	pointwise	icc	-	Promising
khademi2023bardassessment	Khademi (2023)	Education & Grading	pointwise	pct_agreement	-	Mixed
kortemeyer2023physics	Kortemeyer (2023)	Education & Grading	pointwise	pct_agreement	-	Mixed
mizumoto2023aes	Mizumoto & Eguchi (2023)	Education & Grading	pointwise	qwk	0.74	Promising
yancey2023cefr	Yancey et al. (2023)	Education & Grading	pointwise	qwk	-	Mixed
aescomparative2024	Lee et al. (2024)	Education & Grading	pointwise	qwk	0.57	Mixed
floden2024gradingexams	Flodén (2024)	Education & Grading	pointwise	pct_agreement	-	Mixed
grevisse2024asagmed	Grévisse (2024)	Education & Grading	classify	pct_agreement	-	Promising
hou2024classroom	Hou et al. (2024)	Education & Grading	pointwise	icc	-	Mixed
k12shortanswer2024	Henkel et al. (2024)	Education & Grading	classify	kappa	0.70	Promising
koraishi2024langassess	Koraishi (2024)	Education & Grading	pointwise	pct_agreement	-	Mixed
quah2024dentalaes	Quah et al. (2024)	Education & Grading	pointwise	icc	-	Mixed
wang2024beyondagreement	Wang et al. (2024)	Education & Grading	pointwise	qwk	-	Mixed
asagfair2025	Henkel et al. (2025)	Education & Grading	pointwise	kappa	-	Mixed
sefl2025feedback	authors (2025)	Education & Grading	pointwise	f1	-	Promising
studentsjudge2025	Banihashem et al. (2025)	Education & Grading	pointwise	pct_agreement	-	Mixed
k12sci2026	He et al. (2026)	Education & Grading	pointwise	pct_agreement	-	Promising
claimmatch2023	Choi et al. (2023)	Fact-Checking	classify	pct_agreement	-	Promising
ni2024afacta	Ni et al. (2024)	Fact-Checking	classify	pct_agreement	-	Promising
politicaltruths2024	Chatrath et al. (2024)	Fact-Checking	classify	pct_agreement	-	Mixed

Key	Study	Domain	Task	Metric	Agree	Verdict
factsopinions2025	Saju et al. (2025)	Fact-Checking	classify	accuracy	-	Caution
kaikaus2023financecorpora	Kaikaus et al. (2023)	Finance	classify	accuracy	-	Mixed
bizfinbench2025	Lu et al. (2025)	Finance	pointwise	pct_agreement	-	Mixed
trident2025	Hui et al. (2025)	Finance	classify	pct_agreement	-	Mixed
chan2023chateval	Chan et al. (2023)	General NLG	pairwise	pct_agreement	-	Mixed
chiang2023alt	Chiang & Lee (2023)	General NLG	pointwise	pct_agreement	-	Mixed
dubois2023alpacafarm	Dubois et al. (2023)	General NLG	pairwise	winrate_corr	0.98	Promising
hsu2023figurecaption	Hsu et al. (2023)	General NLG	pointwise	spearman	-	Promising
kim2023prometheus	Kim et al. (2023)	General NLG	pointwise	pearson	0.90	Promising
lai2023styletransfer	Lai et al. (2023)	General NLG	pointwise	spearman	-	Mixed
li2023autoj	Li et al. (2023)	General NLG	pairwise	pct_agreement	-	Mixed
wang2023chatgptnlg	Wang et al. (2023)	General NLG	pointwise	spearman	-	Mixed
wang2023pandalm	Wang et al. (2023)	General NLG	pairwise	accuracy	-	Promising
xu2023instructscore	Xu et al. (2023)	General NLG	pointwise	pearson	-	Promising
zhang2023widerdeeper	Zhang et al. (2023)	General NLG	pairwise	pct_agreement	-	Mixed
zheng2023mtbench	Zheng et al. (2023)	General NLG	pairwise	pct_agreement	0.85	Validated
chiang2024arena	Chiang et al. (2024)	General NLG	pairwise	pct_agreement	-	Validated
doostmohammadi2024autoeval	Doostmohammadi et al. (2024)	General NLG	pointwise	pct_agreement	-	Mixed
dubois2024lc	Dubois et al. (2024)	General NLG	pairwise	winrate_corr	0.98	Promising
hashemi2024llmrubric	Hashemi et al. (2024)	General NLG	pointwise	pearson	-	Promising
huang2024nlexpl	Huang et al. (2024)	General NLG	pointwise	pct_agreement	-	Mixed
kim2024prometheus2	Kim et al. (2024)	General NLG	pointwise	pearson	-	Promising
li2024arenahard	Li et al. (2024)	General NLG	pairwise	pct_agreement	0.89	Promising
lin2024wildbench	Lin et al. (2024)	General NLG	pairwise	winrate_corr	0.98	Promising
verga2024poll	Verga et al. (2024)	General NLG	pairwise	pct_agreement	-	Promising
faggioli2023perspectives	Faggioli et al. (2023)	IR & Relevance	classify	kappa	-	Promising
alaofi2024fooledrelevant	Alaofi et al. (2024)	IR & Relevance	classify	pct_agreement	-	Caution
clarke2024cantreplace	Clarke & Dietz (2024)	IR & Relevance	classify	kappa	-	Caution
thomas2024relevance	Thomas et al. (2024)	IR & Relevance	classify	kappa	-	Validated
upadhyay2024largescale	Upadhyay et al. (2024)	IR & Relevance	classify	kappa	-	Promising
zhuang2024beyondyesno	Zhuang et al. (2024)	IR & Relevance	ranking	kendall	-	Promising
arabzadeh2025benchrelevance	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kappa	-	Mixed
arabzadeh2025promptsens	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kappa	-	Mixed
frobe2025assessorsagree	Frobe et al. (2025)	IR & Relevance	classify	kappa	-	Caution
li2023coannotating	Li et al. (2023)	Information Extraction (NER/NLI)	classify	pct_agreement	-	Promising
humanllmcollab2024	Wang et al. (2024)	Information Extraction (NER/NLI)	classify	pct_agreement	-	Promising
kholodna2024activelearning	Kholodna et al. (2024)	Information Extraction (NER/NLI)	classify	f1	-	Promising
kim2024meganno	Kim et al. (2024)	Information Extraction (NER/NLI)	classify	f1	-	Promising
nasution2024chatgptlabel	Nasution & Onan (2024)	Information Extraction (NER/NLI)	classify	accuracy	-	Mixed
ner2024augment	Naraki et al. (2024)	Information Extraction (NER/NLI)	classify	f1	-	Promising
nli2024labeldist	Chen et al. (2024)	Information Extraction (NER/NLI)	classify	pearson	-	Promising
ronningstad2024gptannotator	Ronningstad et al. (2024)	Information Extraction (NER/NLI)	classify	f1	-	Mixed
qiu2025labelensemble	Qiu et al. (2025)	Information Extraction (NER/NLI)	classify	accuracy	-	Promising
zhou2023ifeval	Zhou et al. (2023)	Instruction Following	verify	accuracy	-	Validated
zeng2024llmbar	Zeng et al. (2024)	Instruction Following	pairwise	accuracy	-	Mixed
ifcritic2025	Wen et al. (2025)	Instruction Following	classify	pct_agreement	-	Promising
legaldocrec2025	Pradhan et al. (2025)	Legal	ranking	spearman	0.73	Promising
lemaj2025	Enguehard et al. (2025)	Legal	pointwise	pct_agreement	-	Mixed
kocmi2023gemba	Kocmi & Federmann (2023)	Machine Translation	pointwise	kendall	-	Promising
kocmi2023gembamqm	Kocmi & Federmann (2023)	Machine Translation	verify	pct_agreement	0.96	Validated
lu2023erroranalysis	Lu et al. (2023)	Machine Translation	verify	kendall	-	Promising
sun2023radimpressions	Sun et al. (2023)	Medicine & Clinical	pointwise	pct_agreement	-	Caution
tang2023medevidence	Tang et al. (2023)	Medicine & Clinical	pointwise	pct_agreement	-	Caution
brain2024radgpt	Mitsuyama et al. (2024)	Medicine & Clinical	classify	accuracy	0.94	Mixed

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brake2024clinicalnote	Brake & Schaaf (2024)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
wang2024clinicalevent	Wang et al. (2024)	Medicine & Clinical	classify	pct_agreement	-	Promising
chung2025verifact	Chung et al. (2025)	Medicine & Clinical	verify	pct_agreement	-	Mixed
clinstruct2025	Brake et al. (2025)	Medicine & Clinical	pointwise	icc	0.82	Mixed
croxford2025autoeval	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
croxford2025clinsummjudge	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
healthbench2025	Arora et al. (OpenAI) (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Promising
kim2025emulate	Kim & Kim (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
llmevalmed2025	Zhang et al. (2025)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
szymanski2025limits	Szymanski et al. (2025)	Medicine & Clinical	pairwise	pct_agreement	0.66	Caution
clindialog2026	Sun et al. (2026)	Medicine & Clinical	pointwise	pct_agreement	-	Mixed
frenchmedqa2026	Belmadani et al. (2026)	Medicine & Clinical	pointwise	kappa	-	Mixed
medjudge2026scoping	Li et al. (2026)	Medicine & Clinical	review	nan	-	Caution
counselbench2025	Li et al. (2025)	Mental Health	pointwise	pct_agreement	-	Caution
hua2025mhscoping	Hua et al. (2025)	Mental Health	review	nan	-	Caution
mhtrust2025	Badawi et al. (2025)	Mental Health	pointwise	icc	-	Mixed
mhsupport2026	Badawi et al. (2026)	Mental Health	pointwise	icc	-	Mixed
liu2023trustworthy	Liu et al. (2023)	Methodology & Bias	survey	nan	-	Mixed
saito2023verbositybias	Saito et al. (2023)	Methodology & Bias	pairwise	pct_agreement	-	Caution
wang2023shepherd	Wang et al. (2023)	Methodology & Bias	pointwise	pct_agreement	-	Promising
zhu2023judgelm	Zhu et al. (2023)	Methodology & Bias	pairwise	pct_agreement	-	Promising
bavaresco2024judgebench	Bavaresco et al. (2024)	Methodology & Bias	classify	kappa	0.28	Mixed
chen2024humansorllms	Chen et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
desmond2024evalullm	Desmond et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
fan2024sedareval	Fan et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Promising
feuer2024style	Feuer et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
gu2024survey	Gu et al. (2024)	Methodology & Bias	survey	nan	-	Mixed
jung2024trustescalate	Jung et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Promising
justice2024prejudice	Ye et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
ke2024critiquellm	Ke et al. (2024)	Methodology & Bias	pointwise	spearman	-	Mixed
koo2024cobbler	Koo et al. (2024)	Methodology & Bias	pairwise	winrate_corr	0.44	Caution
li2024compsurvey	Li et al. (2024)	Methodology & Bias	survey	nan	-	Mixed
li2024gen2judge	Li et al. (2024)	Methodology & Bias	survey	nan	-	Mixed
li2024splitmerge	Li et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Mixed
liu2024pairwisepref	Liu et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Promising
pan2024hcdjudge	Pan et al. (2024)	Methodology & Bias	pointwise	nan	-	Mixed
panickssery2024selfpref	Panickssery et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
raina2024robust	Raina et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Unreliable
shankar2024validators	Shankar et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
stureborg2024inconsistent	Stureborg et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Caution
thakur2024judgingjudges	Thakur et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Mixed
wataoka2024selfpref	Wataoka et al. (2024)	Methodology & Bias	pairwise	pct_agreement	-	Caution
zhu2024nationalitybias	Zhu et al. (2024)	Methodology & Bias	pointwise	pct_agreement	-	Caution
calderon2025alttest	Calderon et al. (2025)	Methodology & Bias	classify	pct_agreement	-	Promising
gao2025nlgsurvey	Gao et al. (2025)	Methodology & Bias	survey	nan	-	Mixed
gao2025reranking	Gao et al. (2025)	Methodology & Bias	ranking	kendall	-	Mixed
hu2025trainjudge	Hu et al. (2025)	Methodology & Bias	pointwise	pct_agreement	-	Promising
huang2025empiricaljudge	Huang et al. (2025)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
judgesverdict2025	Han et al. (2025)	Methodology & Bias	pointwise	kappa	-	Promising
li2025prefleakage	Li et al. (2025)	Methodology & Bias	pairwise	pct_agreement	-	Caution
murugadoss2025evaltheeval	Murugadoss et al. (2025)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
voutsa2025biaseddesign	Voutsa et al. (2025)	Methodology & Bias	classify	pct_agreement	-	Caution
yamauchi2025designchoices	Yamauchi et al. (2025)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
imrit2026iaa	James et al. (2026)	Methodology & Bias	review	krippendorff_alpha	-	Mixed

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temperature2026	Li et al. (2026)	Methodology & Bias	pointwise	pct_agreement	-	Mixed
khondaker2023gptaraeval	Khondaker et al. (2023)	Multilingual	classify	f1	-	Caution
mmeval2024	Son et al. (2024)	Multilingual	pairwise	accuracy	-	Mixed
multiljudge2025	Fu et al. (2025)	Multilingual	pointwise	pct_agreement	-	Caution
reliablemulti2026	Zubiaga et al. (2026)	Multilingual	pointwise	pct_agreement	-	Caution
chen2024mllmjudge	Chen et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_agreement	0.79	Mixed
duan2024uimockup	Duan et al. (2024)	Multimodal (Vision-Language)	pointwise	pct_agreement	-	Mixed
lee2024promvision	Lee et al. (2024)	Multimodal (Vision-Language)	pointwise	pearson	-	Mixed
lu2024wildvision	Lu et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_agreement	-	Promising
yasunaga2025mmrewardbench	Yasunaga et al. (2025)	Multimodal (Vision-Language)	pairwise	accuracy	0.72	Caution
chatgptreviewers2024	Ho et al. (2024)	Peer Review & Science	pointwise	pct_agreement	0.07	Caution
liang2024feedback	Liang et al. (2024)	Peer Review & Science	pointwise	pct_agreement	-	Mixed
murad2024quality	Tarakji et al. (2024)	Peer Review & Science	classify	pct_agreement	0.61	Mixed
prismascreen2024	Guo et al. (2024)	Peer Review & Science	classify	kappa	0.91	Validated
rose2025rob	Rose et al. (2025)	Peer Review & Science	classify	kappa	-	Mixed
shopovski2025peerreview	Shopovski et al. (2025)	Peer Review & Science	pointwise	pct_agreement	-	Mixed
guo2023howclose	Guo et al. (2023)	QA & Factuality	pointwise	pct_agreement	-	Mixed
min2023factscore	Min et al. (2023)	QA & Factuality	verify	pct_agreement	-	Promising
li2024pedants	Li et al. (2024)	QA & Factuality	verify	pct_agreement	-	Promising
madaan2024refverdict	Reference-Guided Verdict authors (2024)	QA & Factuality	pointwise	pct_agreement	-	Promising
wei2024safe	Wei et al. (2024)	QA & Factuality	verify	pct_agreement	0.72	Promising
leas2023contentanalysis	Leas et al. (2023)	Qualitative Coding	classify	pct_agreement	-	Mixed
meng2024chalet	Meng et al. (2024)	Qualitative Coding	classify	pct_agreement	-	Mixed
qualigpt2024	Zhang et al. (2024)	Qualitative Coding	classify	pct_agreement	-	Mixed
mehta2025qualgenai	Mehta et al. (2025)	Qualitative Coding	classify	pct_agreement	-	Mixed
qualcoding2025jla	Xiao et al. (2025)	Qualitative Coding	classify	fleiss_kappa	0.46	Mixed
thematic2025charity	Wen et al. (2025)	Qualitative Coding	classify	pct_agreement	-	Mixed
es2023ragas	Es et al. (2023)	RAG	pointwise	pct_agreement	-	Mixed
saadfalcon2023ares	Saad-Falcon et al. (2023)	RAG	classify	accuracy	-	Promising
jin2024ragreward	Jin et al. (2024)	RAG	pairwise	accuracy	-	Mixed
raju2024calc2adj	Raju et al. (2024)	Reasoning & Math	verify	accuracy	-	Mixed
tan2024judgebench	Tan et al. (2024)	Reasoning & Math	pairwise	accuracy	0.55	Caution
mathrobust2026	Yosef et al. (2026)	Reasoning & Math	verify	accuracy	-	Mixed
lee2023rlaif	Lee et al. (2023)	Reward Modeling	pairwise	pct_agreement	-	Mixed
lambert2024rewardbench	Lambert et al. (2024)	Reward Modeling	pairwise	accuracy	-	Mixed
wu2024metarewarding	Wu et al. (2024)	Reward Modeling	pairwise	pct_agreement	-	Mixed
yuan2024selfrewarding	Yuan et al. (2024)	Reward Modeling	pairwise	pct_agreement	-	Mixed
ifreward2026	Wen et al. (2026)	Reward Modeling	ranking	kendall	0.61	Caution
belal2023sentiment	Belal et al. (2023)	Sentiment & Emotion	classify	accuracy	-	Promising
koptyra2023clarinemo	Koptyra et al. (2023)	Sentiment & Emotion	classify	pct_agreement	-	Mixed
zhang2023sentiment	Zhang et al. (2023)	Sentiment & Emotion	classify	accuracy	-	Mixed
niu2024text2emotion	Niu et al. (2024)	Sentiment & Emotion	classify	pct_agreement	-	Mixed
aldeen2023chatgptvshuman	Aldeen et al. (2023)	Social-Science Annotation	classify	accuracy	-	Mixed
cegin2023paraphrases	Cegin et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Mixed
gilardi2023chatgpt	Gilardi et al. (2023)	Social-Science Annotation	classify	accuracy	-	Validated
ollion2023mindhype	Ollion et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Caution
ostyakova2023crowdexperts	Ostyakova et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Mixed
pangakis2023validation	Pangakis et al. (2023)	Social-Science Annotation	classify	f1	0.71	Mixed
reiss2023testing	Reiss (2023)	Social-Science Annotation	classify	pct_agreement	-	Caution
wu2023llmworkers	Wu et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Mixed
zhu2023reproduce	Zhu et al. (2023)	Social-Science Annotation	classify	pct_agreement	-	Mixed
li2024stance	Li & Conrad (2024)	Social-Science Annotation	classify	pct_agreement	-	Mixed
pangakis2024keephumans	Pangakis & Wolken (2024)	Social-Science Annotation	classify	f1	-	Promising
ziems2024css	Ziems et al. (2024)	Social-Science Annotation	classify	accuracy	-	Mixed

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ahmad2025moralmachine	Ahmad & Takemoto (2025)	Social-Science Annotation	classify	pct_agreement	-	Mixed
horych2025promisespitfalls	Horych et al. (2025)	Social-Science Annotation	classify	f1	-	Mixed
tornberg2025political	Törnberg (2025)	Social-Science Annotation	classify	accuracy	-	Validated
gao2023humanlike	Gao et al. (2023)	Summarization	pointwise	spearman	-	Mixed
liu2023geval	Liu et al. (2023)	Summarization	pointwise	spearman	0.51	Promising
luo2023factinconsistency	Luo et al. (2023)	Summarization	verify	pct_agreement	-	Mixed
shen2023notyet	Shen et al. (2023)	Summarization	pointwise	spearman	-	Caution
lee2024unisumeval	Lee et al. (2024)	Summarization	pointwise	pct_agreement	-	Promising
song2024finesure	Song et al. (2024)	Summarization	verify	pct_agreement	-	Promising
tang2024tofueval	Tang et al. (2024)	Summarization	verify	pct_agreement	-	Caution
zhang2024newssumm	Zhang et al. (2024)	Summarization	pointwise	pct_agreement	-	Promising
wu2023hpsv2	Wu et al. (2023)	Text-to-Image	pointwise	accuracy	-	Promising
lin2024vqascore	Lin et al. (2024)	Text-to-Image	pointwise	pearson	-	Promising